



VPRO SKILLS

FAQ / INTERVIEW PREPARATION

Pandas Interview Questions & Answers

STUDENT LEARNING RESOURCE

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How to Use This Guide

This handbook brings together 50 curated questions and answers on Pandas, organized into a single quick-reference list for focused revision. Each question is numbered for easy reference, and answers are broken down into definitions, examples, code, and comparison tables wherever the source material provides them.

- Use the Contents page to jump straight to a section.
- Code blocks are shown in a monospaced, shaded panel — read them alongside the explanation above.
- Comparison tables summarize key differences at a glance; use them for quick revision before an interview.
- The Quick Revision page at the end lists every question so you can self-test your recall.

QUESTIONS

Q01. What is Pandas?

Pandas is a Python library used to work with data.

It helps us:

- Read data from CSV, Excel, JSON, SQL.
- Clean data.
- Analyze data.
- Filter, group, merge, and transform data.

INTERVIEW TIP

Say "Pandas is mainly used for data manipulation and data analysis."

Q02. What is a Series?

A Series is a one-dimensional data structure.

It contains one column of data with an index.

```
import pandas as pd
s = pd.Series([10,20,30])
```

Output

Index	Value
0	10
1	20
2	30

Q03. What is a DataFrame?

A DataFrame is a two-dimensional table.

It has rows and columns.

```
df = pd.DataFrame({
    "Name": ["Ram", "Sam"],
    "Salary": [50000, 60000]
})
```

It looks like an Excel table.

Q04. Difference between Series and DataFrame?**Q05. How do you read a CSV file?**

Use `read_csv()`.

```
df = pd.read_csv("employees.csv")
```

This reads the CSV into a DataFrame.

Q06. How do you read an Excel file?

Use `read_excel()`.

```
df = pd.read_excel("employees.xlsx")
```

Q07. How do you check the first 5 rows?

Use `head()`.

```
df.head()
```

Useful to quickly see sample data.

Q08. How do you check the last 5 rows?

Use `tail()`.

```
df.tail()
```

Q09. How do you know the number of rows and columns?

Use `shape`.

```
df.shape
```

Example output:

```
(100,5)
```

Means 100 rows and 5 columns.

Q10. How do you know column names?

```
df.columns
```

Returns all column names.

Q11. How do you know data types?

```
df.dtypes
```

Shows whether columns are int, float, object, datetime, etc.

Q12. What does info() do?

```
df.info()
```

Shows:

- Number of rows.
- Column names.
- Data types.
- Missing values.
- Memory usage.

Very useful during data cleaning.

Q13. What does describe() do?

```
df.describe()
```

Shows statistics of numeric columns:

- Count
- Mean
- Minimum
- Maximum
- Standard deviation
- Quartiles

Q14. How do you select one column?

```
df["Salary"]
```

Returns a Series.

Q15. How do you select multiple columns?

```
df[["Name", "Salary"]]
```

Returns a DataFrame.

Q16. What is loc?

loc selects data using row labels and column names.

```
df.loc[0]
```

Returns first row.

EXAMPLE

```
df.loc[0, "Salary"]
```

Q17. What is iloc?

iloc selects data using row position and column position.

```
df.iloc[0]
```

Returns first row.

EXAMPLE

```
df.iloc[0,1]
```

Returns first row, second column.

Q18. Difference between loc and iloc?

Q19. How do you filter rows?

```
df[df["Salary"] > 50000]
```

Returns employees with salary above 50,000.

Q20. How do you filter multiple conditions?

```
df[(df["Salary"] > 50000) & (df["Department"] == "IT")]
```

Use & for AND.

Use | for OR.

Q21. How do you rename columns?

```
df.rename(columns={"Salary": "CTC"})
```

Rename one or many columns.

Q22. How do you change data type?

```
df["Salary"] = df["Salary"].astype(float)
```

Changes integer to float.

Q23. How do you add a new column?

```
df["Bonus"] = df["Salary"] * 0.10
```

Creates a new column.

Q24. How do you update a column?

```
df["Salary"] = df["Salary"] + 5000
```

Updates all salary values.

Q25. How do you delete a column?

```
df.drop("Bonus", axis=1)
```

axis=1 means column.

Q26. How do you delete rows?

```
df.drop(2)
```

Deletes row with index 2.

Q27. What is drop()?

drop() removes rows or columns.

EXAMPLES

```
df.drop(1)  
df.drop("Salary", axis=1)
```

Q28. How do you find missing values?

```
df.isnull()
```

Returns True where value is missing.

Q29. How do you count missing values?

```
df.isnull().sum()
```

Shows missing value count in every column.

Q30. How do you remove missing values?

```
df.dropna()
```

Removes rows containing missing values.

Q31. How do you fill missing values?

```
df.fillna(0)
```

Replaces missing values with 0.

Q32. What is Forward Fill (ffill)?

Copies previous value.

```
df.fillna(method="ffill")
```

Useful in time-series data.

Q33. What is Backward Fill (bfill)?

Copies next available value.

```
df.fillna(method="bfill")
```

Q34. How do you remove duplicate rows?

```
df.drop_duplicates()
```

Removes duplicate records.

Q35. How do you find duplicate rows?

```
df.duplicated()
```

Returns True for duplicates.

Q36. What is groupby()?

groupby() groups rows with the same value.

EXAMPLE

```
df.groupby("Department")["Salary"].mean()
```

Find average salary department-wise.

Q37. How do you calculate average salary by department?

```
df.groupby("Department")["Salary"].mean()
```

Returns department-wise average salary.

Q38. What is agg()?

agg() performs multiple calculations together.

```
df.groupby("Department").agg({  
    "Salary": ["mean", "max", "min"]  
})
```

Returns average, maximum, and minimum salary.

Q39. Difference between agg() and transform()?

EXAMPLE

```
df["DeptAvg"] = df.groupby("Department")["Salary"].transform("mean")
```

Q40. What is merge()?

merge() joins two DataFrames like SQL JOIN.

```
pd.merge(df1, df2, on="EmployeeID")
```

Q41. Types of Merge?

Q42. What is concat()?

concat() combines DataFrames.

Rows:

```
pd.concat([df1, df2])
```

Columns:

```
pd.concat([df1, df2], axis=1)
```

Q43. Difference between merge() and concat()?

Q44. What is pivot_table()?

Creates a summary table.

```
pd.pivot_table(  
    df,  
    values="Sales",  
    index="Region",  
    columns="Year",  
    aggfunc="sum"  
)
```

Useful for reports.

Q45. What is apply()?

apply() applies a function to rows or columns.

```
df["Salary"] = df["Salary"].apply(lambda x: x * 1.10)
```

Increases salary by 10%.

Q46. Why is vectorization faster than apply()?

Vectorization performs operations on the entire column at once.

EXAMPLE

```
df["Salary"] = df["Salary"] * 1.10
```

It is faster and uses less memory than looping through rows.

Q47. How do you convert a column to DateTime?

```
df["Date"] = pd.to_datetime(df["Date"])
```

Now Pandas understands it as a date.

Q48. How do you extract Year, Month, and Day?

```
df["Year"] = df["Date"].dt.year  
df["Month"] = df["Date"].dt.month  
df["Day"] = df["Date"].dt.day
```

Useful for reports and dashboards.

Q49. How do you optimize Pandas for large CSV files?

Use chunk processing.

```
for chunk in pd.read_csv("sales.csv", chunksize=100000):  
    print(chunk.head())
```

Benefits:

- Reads small chunks.
- Saves memory.
- Works with very large files.

Q50. What are the top Pandas interview tips for experienced candidates?

In interviews, explain why you use a method.

EXAMPLE

- Use `groupby()` for department-wise calculations.
- Use `transform()` when you need results back in every row.
- Use `merge()` to combine customer and order tables.
- Use vectorized operations instead of loops for better performance.
- Handle missing values before analysis.
- Use `loc` for updating values safely.

Quick Revision

Scan through these questions to self-test your recall before an interview.

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- Q02. What is a Series?
- Q03. What is a DataFrame?
- Q04. Difference between Series and DataFrame?
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- Q21. How do you rename columns?
- Q22. How do you change data type?
- Q23. How do you add a new column?
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- Q27. What is drop()?
- Q28. How do you find missing values?
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- Q32. What is Forward Fill (ffill)?
- Q33. What is Backward Fill (bfill)?
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- Q35. How do you find duplicate rows?
- Q36. What is groupby()?
- Q37. How do you calculate average salary by department?
- Q38. What is agg()?
- Q39. Difference between agg() and transform()?
- Q40. What is merge()?
- Q41. Types of Merge?
- Q42. What is concat()?
- Q43. Difference between merge() and concat()?

- Q44. What is `pivot_table()`?
- Q45. What is `apply()`?
- Q46. Why is vectorization faster than `apply()`?
- Q47. How do you convert a column to `DateTime`?
- Q48. How do you extract Year, Month, and Day?
- Q49. How do you optimize Pandas for large CSV files?
- Q50. What are the top Pandas interview tips for experienced candidates?